

Learn LaTeX

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June 8, 2026

1 Introduction

This is a simple LaTeX document to demonstrate how to use LaTeX for type-setting mathematical equations and creating structured documents.

2 Writing Formulas

Let's begin with a formula $e^{i\pi} + 1 = 0$, which is known as Euler's identity. It beautifully connects five fundamental mathematical constants: e , i , π , 1 , and 0 .

$$\int_a^b f(x) dx = F(b) - F(a)$$

1. We can do another:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

2. We can also write matrices:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

3. We can also use continuous fractions:

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}}$$

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3 More Formulas

- Integral:

$$\int_a^b f(x) dx = F(b) - F(a)$$

- Triple integral:

$$\iiint_a^b \int_c^d f(x, y) dx dy$$

- Vector:

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

- Limit:

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

- Matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

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